

DW_lp_cntr_updn_df

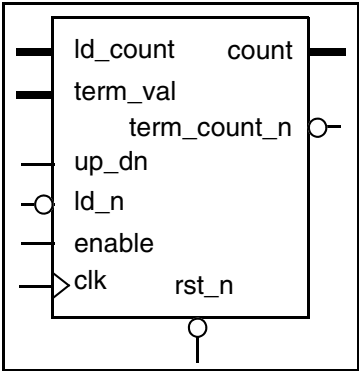
Low Power Counter with Dynamic Terminal Count Flag

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Reduced power through efficient clock gating
- Parameter-controlled counter width
- Dynamically controlled terminal count flag specification
- Parameter-controlled selection of registered flag output
- Parameter-controlled selection of synchronous versus asynchronous reset
- Synchronous enable
- Synchronous load

Revision History



Description

DW_lp_cntr_updn_df is a general-purpose up/down counter with dynamic count-to logic.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
clk	1 bit	Input	Clock
rst_n	1 bit	Input	Reset input (active low)
enable	1 bit	Input	Enable input
up_dn	1 bit	Input	Up versus down control input (1 = up, 0 = down)
ld_n	1 bit	Input	Counter load control (active low)
ld_count	<i>width</i> bits	Input	Counter load value
term_val	<i>width</i> bits	Input	Terminal count value
count	<i>width</i> bits	Output	Output count bus
term_count_n	1 bit	Output	Terminal count (active low)

Table 1-2 Parameter Description

Parameter	Values	Description
width	1 to 2048	Word length of counter
rst_mode	0 or 1 Default: 0	Reset mode 0 = Asynchronous reset 1 = Synchronous reset
reg_trmcnt	0 or 1 Default: 0	Register term_count_n output 0 = Not registered 1 = Registered

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
rtl	Synthesis model	- DesignWare (P-2019.03 and later) - DesignWare-LP^a (before P-2019.03)

- a. For versions before P-2019.03, you must enable minPower as follows:
`set_synthetic_library {dw_foundation.sldb dw_minpower.sldb}`

Table 1-4 Simulation Models

Model	Function
DW03.DW_LP_CNTR_UPDN_DF_CFG_SIM	Design unit name for VHDL simulation
dw/dw03/src/DW_lp_cntr_updn_df_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_lp_cntr_updn_df.v	Verilog simulation model source code

Table 1-5 Counter Operation Truth Table

rst_n	ld_n	enable	up_dn	Operation
0	X	X	X	Reset
1	0	X	X	Load
1	1	0	X	Hold (disabled)
1	1	1	0	Count down
1	1	1	1	Count up

When the count value equals the value on the `term_val` port, the signal `term_count_n` (terminal count) is asserted (low). The signal `term_count_n` can be connected directly to `ld_n` to synchronously reset the counter to a predefined value on the input pin of the data bus, `ld_count`.

The counter is *width* bits wide and has 2^{width} states from “000...0” to “111...1”. The counter is clocked on the positive edge of *clk*.

The reset signal, *rst_n*, is active low and provides for an asynchronous reset of the counter to “000...0”. If the reset pin is connected to ‘1’, then the reset logic is not synthesized, resulting in a smaller and faster counter.

The *term_val* is an input bus that ranges from 0 to *width* - 1. When the counter output, *count*, equals *term_val*, *term_count_n* goes active (low) for one clock cycle.

The *up_dn* input controls whether the counter counts up (*up_dn* is high) or down (*up_dn* is low), starting on the next positive edge of *clk*.

The counter is loaded with *data* by asserting *ld_n* (low) and applying data to *ld_count*. The data load operation is synchronous with respect to the positive edge of *clk*.

The count enable pin, *enable*, is active high. When *enable* is high, the counter is active. When *enable* is low, the counter is disabled, and *count* remains at the same value.

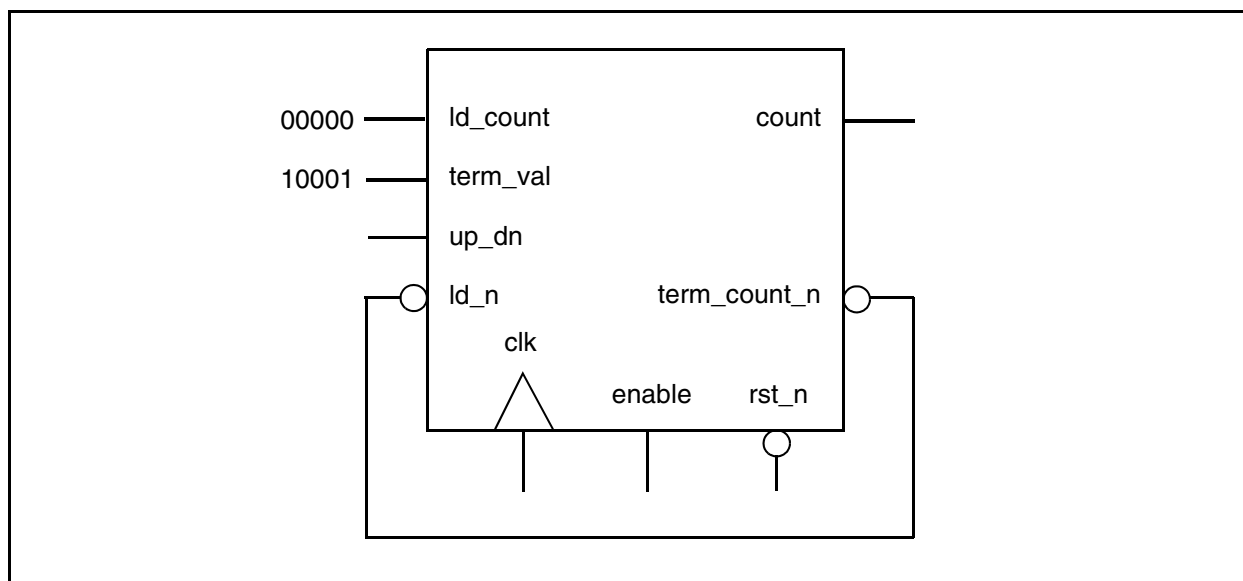
Application Example

An example application of DW_lp_cntr_updn_df is to count from 0 to 17, repeatedly. This is done by:

1. Connecting the terminal count output signal, *term_count_n*, to the *ld_n* input.
2. Setting the *ld_count* input to the start of the count sequence (for example, “00000”).
3. Connecting the *term_val* input to the end of the count sequence (for example, “10001”).

The count output then cycles through the states 00000 (*ld_count*) to 10001 (*term_val*); see [Figure 1-1](#).

Figure 1-1 Counter Application: *width* = 5



Timing Diagrams

The following figures show various timing diagrams for DW_lp_cntr_updn_df.

Figure 1-2 Functional Operation: Reset, Load, and Count-to Sequence

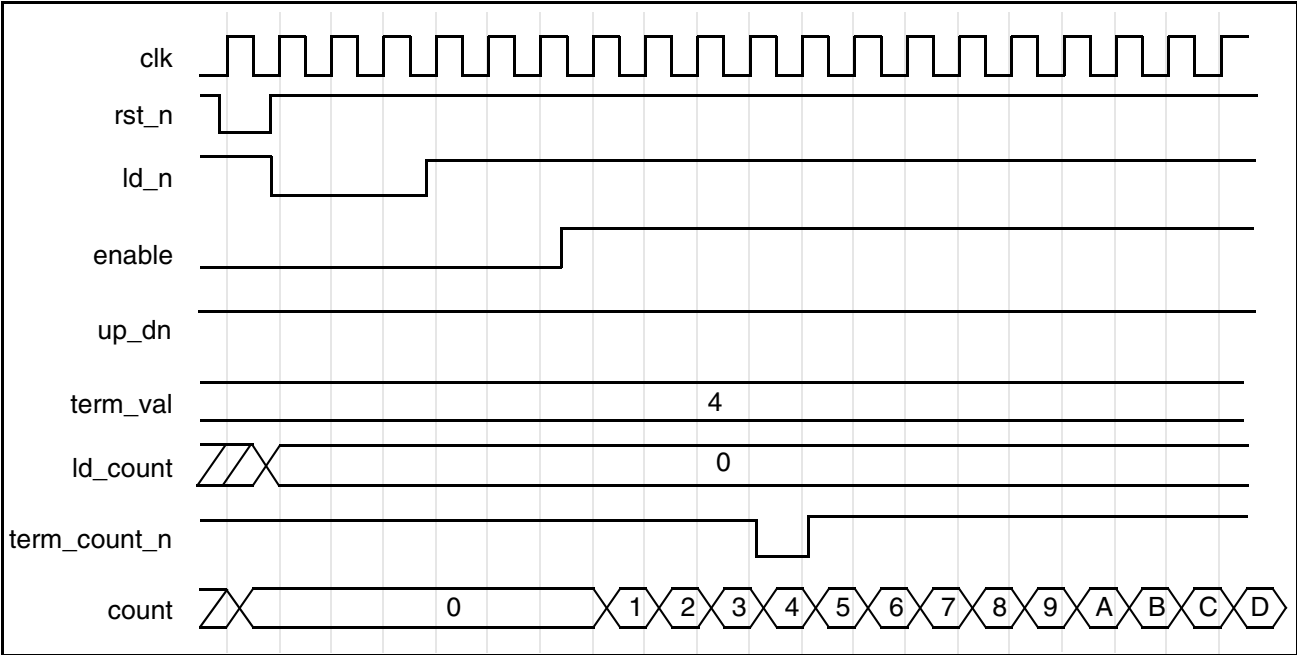
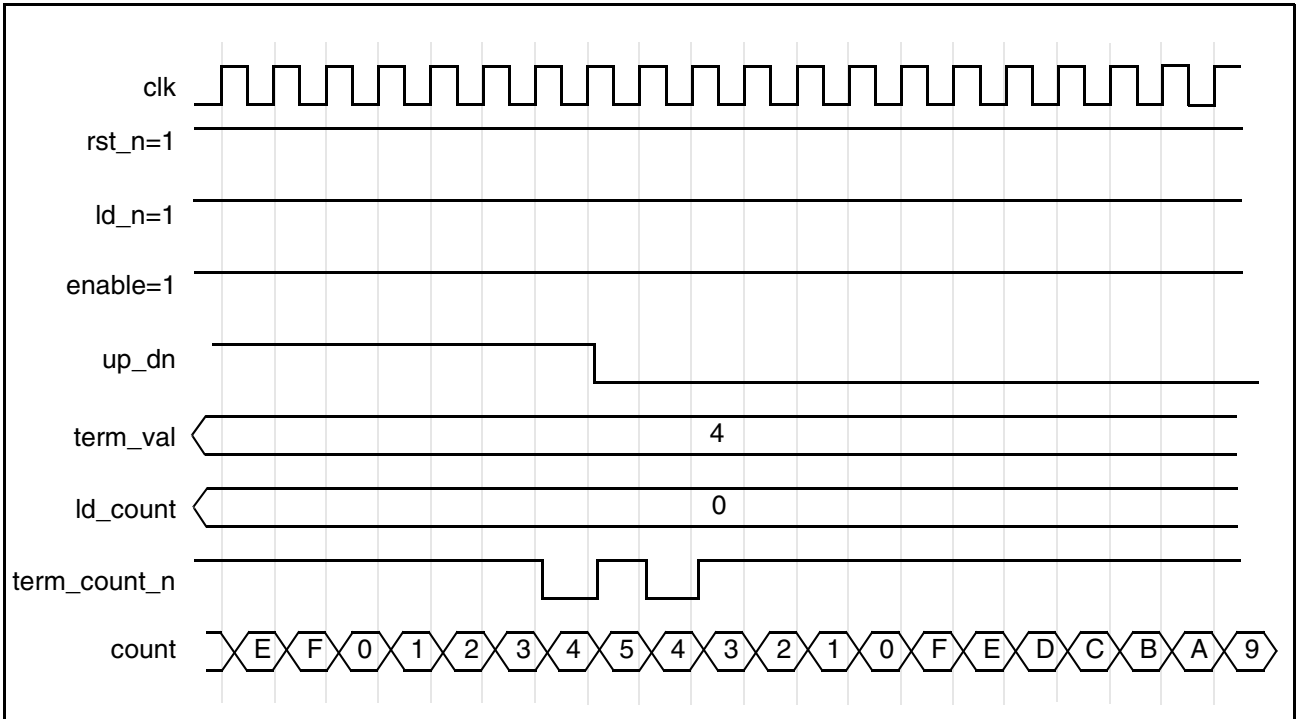


Figure 1-3 Functional Operation: Up and Down Counting, and Count-to Sequence



Related Topics

- [Logic – Sequential Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_Foundation_comp.all;

entity DW_lp_cntr_updn_df_inst is
    generic (
        inst_width : POSITIVE := 8;
        inst_rst_mode : NATURAL := 0;
        inst_reg_trmcnt : NATURAL := 0
    );
    port (
        inst_clk : in std_logic;
        inst_rst_n : in std_logic;
        inst_enable : in std_logic;
        inst_up_dn : in std_logic;
        inst_ld_n : in std_logic;
        inst_ld_count : in std_logic_vector(inst_width-1 downto 0);
        inst_term_val : in std_logic_vector(inst_width-1 downto 0);
        count_inst : out std_logic_vector(inst_width-1 downto 0);
        term_count_n_inst : out std_logic
    );
end DW_lp_cntr_updn_df_inst;

architecture inst of DW_lp_cntr_updn_df_inst is

begin

    -- Instance of DW_lp_cntr_updn_df
    U1 : DW_lp_cntr_updn_df
        generic map (width => inst_width,
                    rst_mode => inst_rst_mode,
                    reg_trmcnt => inst_reg_trmcnt )

        port map ( clk => inst_clk, rst_n => inst_rst_n,
                  enable => inst_enable, up_dn => inst_up_dn,
                  ld_n => inst_ld_n, ld_count => inst_ld_count,
                  term_val => inst_term_val,
                  count => count_inst, term_count_n => term_count_n_inst );

end inst;
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_lp_cntr_updn_df_inst(
    inst_clk, inst_rst_n,
    inst_enable, inst_up_dn,
    inst_ld_n, inst_ld_count,
    inst_term_val,
    count_inst, term_count_n_inst );

parameter width = 8;
parameter rst_mode = 0;
parameter reg_trmcnt = 0;

input inst_clk;
input inst_rst_n;
input inst_enable;
input inst_up_dn;
input inst_ld_n;
input [width-1 : 0] inst_ld_count;
input [width-1 : 0] inst_term_val;
output [width-1 : 0] count_inst;
output term_count_n_inst;

// Instance of DW_lp_cntr_updn_df
DW_lp_cntr_updn_df #(width, rst_mode, reg_trmcnt)
    U1 (.clk(inst_clk), .rst_n(inst_rst_n),
        .enable(inst_enable), .up_dn(inst_up_dn),
        .ld_n(inst_ld_n), .ld_count(inst_ld_count),
        .term_val(inst_term_val),
        .count(count_inst), .term_count_n(term_count_n_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
March 2019	DWBB_201903.0	■ Clarified some information about minPower in Table 1-3 on page 2 ■ Added this Revision History table and the document links on this page

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